

EXPERIMENT -07

AIM- To perform Simple Linear Regression with R

CODE-

```
import numpy as np import matplotlib.pyplot as plt from
sklearn.datasets import load_diabetes from sklearn.linear_model
import LinearRegression from sklearn.model_selection import
train_test_split from sklearn.metrics import
mean_squared_error,r2_score
```

```
#step 1 :Load the DataSet print("Loading
Dataset...") diabetes=load_diabetes()
```

```
#use only one features (BMI) to make it simple regression
x=diabetes.data[:,2].reshape(-1,1) y=diabetes.target
```

```
print(f"Dataset loaded with {len(x)} samples")
```

```
#split data into trianing and testing
```

```
x_train,x_test,y_train,y_test= train_test_split( x,y
,test_size=0.2,random_state=42
)
```

```
#step-3: CREATE AND TRAIN MODEL
```

```
model=LinearRegression() model.fit(x_train,y_train)
print("\n__ Model Training Complete__")
```

```

print(f"Slope (coefficient):{model.coef_[0]:.2f}")
print(f"Intercept:{model.intercept_:.2f}")

#step-4 Make Predictions y_pred=model.predict(x_test)

#step 5: Evaluate the model
mse=mean_squared_error(y_test,y_pred) r2=r2_score(y_test,y_pred)

print("\n ---Model Performance---")
print(f"Mean Squared Error:{mse:.2f}") print(f"R2
Squared :{r2:.2f}")

#visulisation the results plt.figure(figsize=(10,6))

#plot training data
plt.scatter(x_train,y_train,color='blue',alpha=0.5,label="training Data")

#plot test data
plt.scatter(x_test,y_test,color='Green',alpha=0.5,label="test Data")

#plot regression line
plt.plot(x_test,y_pred,color='red',linewidth=2,alpha=0.5,label="regressio n
line")

plt.xlabel('BMI (Body Mass Index)') plt.ylabel('Disease
Prograssion') plt.title("Simple Linear Regression :BMI Vs Disease
Prograssion") plt.legend() plt.grid(True,alpha=0.3) plt.show

```

OUTPUT

Loading Dataset...
Dataset loaded with 442 samples

__ Model Training Complete __
Slope (coefficient):998.58
Intercept:152.00

---Model Performance---
Mean Squared Error:4061.83
R2 Squared :0.23

[3]: <function matplotlib.pyplot.show(close=None, block=None)>

